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Combining transfer learning and constrained long short-term memory for power generation forecasting of newly-constructed photovoltaic plants

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ABSTRACT

Photovoltaic power generation (PVPG) forecasting has attracted increasing research and industry attention due to its significance for energy management, infrastructure planning, and budgeting. Emerging deep learning (DL) models based on historical data have provided effective solutions for PVPG forecasting with great success. However, newly-constructed photovoltaic (NCPV) plants often lack collections of historical data, and thus it is difficult to forecast their future generation accurately. In this work, combining transfer learning (TL) and DL models, we initially propose two parameter-transferring strategies and a constrained long short-term memory (C-LSTM) model, to address the hourly day-ahead PVPG forecasting problem of NCPV plants. The K-nearest neighbors (KNN) algorithm is utilized to extract prior knowledge as physical constraints, which can guide the training process of C-LSTM. The performances of different TL methods combined with C-LSTM are evaluated specifically, and appropriate ones are determined accordingly. The proposed models are evaluated based on real-life datasets collected from actual PV plants in Australia. The results demonstrate that the proposed C-LSTM model outperforms the standard LSTM model with higher forecasting accuracy. In addition, the results also indicate that significant improvements in forecasting accuracy and stability can be obtained by the proposed TL strategies combined with C-LSTM, regardless of different sky conditions (i.e., clear sky, partly cloudy sky, and overcast sky), compared to the conventional machine learning and statistical models in the literature. The forecasting skill of the combined model has improved up to 68.4% compared with the reference persistence model.

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1. Introduction

In the last decades, an increasing amount of electricity has been generated from renewable energy sources, among which photovoltaic (PV) energy is a major source, with a high penetration rate in the market [1,2]. According to a report from the International Renewable Energy Agency (IRENA), the installed capacity from PV power has increased dramatically from 40 GW in 2010 to 578 GW in 2019, and global installed PV power capacity is estimated to reach 1700 GW by 2030 [3]. One particular characteristic of PV energy is that the output of a power plant depends largely on weather conditions. Consequently, the PV power generation (PVPG) of a system

Abbreviations: Adam, Adaptive moment estimation; ARIMA, Auto-regressive integrated moving average; C-LSTM, Constrained long short-term memory; CNN, Convolutional neural network; DL, Deep learning; ECMWF, European center for medium-range weather forecasts; FCNN, Fully-connected neural network; IRENA, International renewable energy agency; KNN, K-nearest neighbors; LSTM, Long short-term memory; ML, Machine learning; MSE, Mean square error; NCPV, Newly-constructed photovoltaic; NRMSE, Normalized root mean square error; NWP, Numerical weather prediction; PCC, Pearson correlation coefficient; PSO, Particle swarm optimization; PV, Photovoltaic; PVPG, PV power generation; RMSE, Root mean square error; RNN, Recurrent neural network; SD, Standard deviation; SGD, Stochastic gradient descent; SL, Significance level; TL, Transfer learning.

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is highly uncertain and varies dynamically with time, since the main controlling factors, including the intensity of solar radiation and a variety of meteorological variables, are usually both uncertain and ungovernable [4]. However, integration of uncertain PV power into the grid may bring great challenges to the safe and stable operation of the power system. Therefore, an accurate and stable PVPG forecast is highly desired when integrating PV power into the grid, to ensure the stability and reliability of the power system.

The issue of PVPG forecasting has attracted increasing focus from researchers, and a substantial amount of research has been conducted on it. The forecasting methods illustrated in the literature can be broadly classified into four typical categories [5]: the physical model [6,7]; the statistical model [8–11]; the machine learning (ML) model [12–16]; and the hybrid model [17–21]. Among them, deep learning (DL) [22], a sub-field of ML, has been gaining popularity with the development of smart metering techniques in recent years. DL models possess strong predictive capabilities in terms of learning feature representations and modeling complex relationships commonly hidden in big data. Although various DL models, such as fully-connected neural networks (FCNN) and convolutional neural networks (CNN), have been applied for PVPG forecasting [16,17,21], these approaches are designed to only simulate point-to-point mapping between the input space and the output space. However, time-dependence is not considered by FCNN or CNN, whereas it exists in time-series PV data. As a result, a better choice is the recurrent neural network (RNN) [23], a class of DL models, in which an internal self-looped structure exists, allowing the network to preserve previous information.

In the RNN class, the long short-term memory (LSTM) is one of the most popular representatives, and has been widely adopted in time-series regression problems. Compared to conventional RNNs, LSTM can not only handle the correlation of time-series data, but also prevent gradient vanishing and gradient exploding, which are found within conventional RNNs. Consequently, LSTM has demonstrated great successes in PVPG forecasting in practice. For example, Zheng et al. [24] proposed the LSTM model incorporated with the particle swarm optimization (PSO) algorithm to forecast multi-region solar power outputs. The PSO algorithm is used to optimize the parameters of LSTM. In addition, Gensler et al. [25] proposed the LSTM model combined with an auto-encoder (Auto-LSTM) to solve the problem of solar power forecasting. The results using Auto-LSTM showed superior performance compared to ANN, as well as other reference models. Moreover, Nasser and Mahmoud [26] applied the LSTM model to accurately forecast the output power of PV systems based on hourly datasets of different sites for a year. The obtained results demonstrated the superiority of LSTM over other compared models.

DL techniques, especially LSTM, greatly advanced PVPG forecasting and improved its accuracy; however, several limitations remain in PVPG forecasting in previous works. Firstly, although numerous excellent DL models have been proposed, most of them are data-consuming and designed based on the assumption that sufficient training data and future data under the same distribution are available. However, in real-world situations, such as PVPG forecasting for some newly-constructed PV (NCPV) plants, this assumption may not hold any longer. For NCPV plants, sizable and long-term training data are challenging to obtain, thus creating certain difficulties in maintaining the accuracy of DL models. As a consequence, determination of how to improve forecasting performances of newly-constructed PV plants with limited training data is a significant issue that needs to be solved urgently. In such cases, knowledge transfers or transfer learning between task domains would be desirable.

Transfer learning (TL) is a viable technique in ML that focuses on promoting the capability of solving a specific problem (target domain) by using knowledge extracted from different but related problems (source domains) when the target domain has fewer high-quality training data [27]. In recent years, TL techniques have been broadly adopted in many research fields, such as image segmentation, classification, and regression. For example, Opbroek et al. [28] proposed four classifiers based on TL that can train a classification scheme with limited training data, in addition to a larger amount of other training data with slightly dissimilar characteristics. Their results showed that TL can assist to reduce classification errors by up to 60%. Additionally, Allard et al. [29] utilized TL with three DL networks for hand-gesture signal classification. The proposed TL scheme was demonstrated to significantly enhance performance for all three networks on the two public datasets, achieving an off-line accuracy of 98.31%. In terms of regression, Cai et al. [27] proposed a two-layer TL architecture to improve the forecasting accuracy of short-term load. Case studies showed that their proposed TL architecture always improved the forecasting accuracy of classic load forecasting algorithms, especially when the load data of the target domain are insufficient. Furthermore, other applications of TL can be found in Refs. [30–33]. Motivated by the successes obtained in previous works, TL could constitute a potential solution for the PVPG forecasting problem of NCPV plants or PV plants without a sufficient record of data, and thus is being studied in this work.

Secondly, recent DL models are applied based only on massive data, and no prior knowledge is involved in the construction of the model. For the standard LSTM, however, it may produce physically unreasonable predictions during training and forecasting processes [24–26], particularly for the case of the NCPV plant which has limited training data. This may occur in a deep neural network without incorporating some prior knowledge of a specific engineering problem [34]. Physical constraints derived from prior knowledge of PV are required to assist the training of DL models. In Ref. [5], some physical constraints were extracted from domain knowledge of PV based on sufficient historical data, and excellent performance was obtained by their proposed DL framework. However, the way of constructing constraints is not suitable for PVPG forecasting of NCPV plants with limited training data in target domains. As a result, identification of how to extract physical constraints from prior knowledge for NCPV plants and design constrained DL models remains challenging and is worthy of investigation.

Considering the above limitations in previous works, much room exists to improve DL models in terms of predictability, reliability, and interpretability. Consequently, in this work, combining transfer learning (TL) and DL models, we propose two parameter-transferring strategies and a constrained long short-term memory (C-LSTM) model as the transferable framework, to address the hourly day-ahead PVPG forecasting problem of NCPV plants. The main novelties and contributions of our work are summarized as follows:

(1) Two parameter-transferring strategies are proposed based on C-LSTM for the day-ahead PVPG forecasting problem of NCPV plants. Specific TL methods incorporated with C-LSTM are evaluated specifically, and appropriate ones are determined accordingly.
(2) The C-LSTM model is initially proposed by using the K-nearest neighbors (KNN) algorithm for the first time to extract prior knowledge as constraints that can be integrated into the model construction and guide the training process of C-LSTM.
(3) The proposed C-LSTM model outperforms the standard LSTM model with higher forecasting accuracy, which shows the

significance of considering prior knowledge into the construction of DL models.

(4) The proposed TL strategies combined with C-LSTM can obtain significant improvements in terms of forecasting accuracy and stability, regardless of different sky conditions, compared to the conventional machine learning and statistical models in the literature.

The remainder of this paper is organized as follows. In Section 2, the definition of transfer learning in PVPG forecasting is introduced and TL strategies are then proposed. Subsequently, the C-LSTM model is specifically illustrated as a transferable framework in Section 3. The description of adopted data is reported in Section 4. In section 5, the proposed approaches are evaluated specifically. The performances of different forecasting models in the literature are also compared. Finally, the paper is concluded in Section 6.

2. Methodology of transfer learning

The methodology of transfer learning is introduced in Section 2. Specifically, the definition of transfer learning in PVPG forecasting is illustrated in Subsection 2.1. Subsequently, two parameter-transferring strategies with specific transfer methods are proposed in Subsection 2.2.

2.1. Transfer learning definition in PVPG forecasting

As shown in Fig. 1, traditional ML models attempt to learn tasks individually based only on the corresponding learning domain. However, TL focuses on promoting the capability of solving a specific problem (target domain) by using knowledge extracted from different but related problems (source domains), when the target domain has fewer high-quality training data [27].

Therefore, the definition of TL can be proposed as: given a source domain $\mathcal{D}^S$ and learning task $\mathcal{T}^S$, a target domain $\mathcal{D}^T$ and learning task $\mathcal{T}^T$, transfer learning aims to improve the learning of the target predictive function $f_T(\cdot)$ in $\mathcal{D}^T$ using the knowledge in $\mathcal{D}^S$ and $\mathcal{T}^S$, where $\mathcal{D}^S \neq \mathcal{D}^T$ and $\mathcal{T}^S = \mathcal{T}^T$.

In this work, given domains $\mathcal{D}^S = \{\mathcal{X}^S, P^S(X)\}$ and $\mathcal{D}^T = \{\mathcal{X}^T, P^T(X)\}$, since both domains have same feature terms, the condition $\mathcal{D}^S \neq \mathcal{D}^T$ implies $P^S(X) \neq P^T(X)$. In terms of PVPG forecasting, this means that both domains possess the same feature terms (e.g., meteorological variables, such as solar radiation, temperature, humidity, and wind speed) as inputs, whereas the marginal probability distributions of the above feature terms are different. In addition, the condition $\mathcal{T}^S = \mathcal{T}^T$ indicates that both domains share the same task. This implies that the predictive function $f_S(\cdot)$ learned from the source domain can be utilized in the target domain after careful fine-tuning, and then $f_S(\cdot) \rightarrow f_T(\cdot)$. The learning task for both domains is to acquire accurate mapping relationships between a variety of meteorological variables and the output of the PV system.

2.2. Proposed transfer learning strategies

Adhering to the transfer learning definition, parameter-transferring approaches which assume that individual models for related tasks should share some model parameters, are proposed in Subsection 2.2. Therefore, the knowledge learned from the source domain can be transferred to the target domain by different parameter-sharing methods. Specifically, a basic DL model is pre-trained in the source domain at first. Subsequently, parameters of the pre-trained model are fully or partially transferred to the model in the target domain for further fine-tuning by appropriate TL methods. As a consequence, two specific TL strategies, termed shallow transferring and deep transferring, are proposed in this study to assist DL models to address the PVPG forecasting problem of newly-constructed PV plants. The principles of the proposed TL strategies are visually displayed in Fig. 2 and their corresponding definitions are proposed as follows.

2.2.1. Strategy #1 (shallow transferring)

Transfer the inner parameters of different layers of the pre-trained model in the source domain, to the model in the target domain as the parameter initialization, and then fine-tune the entire parameters of the new model in the target domain.

2.2.2. Strategy #2 (deep transferring)

Transfer the entire parameters of the pre-trained model in the source domain to the model in the target domain, freeze the parameters in some layers of the new model, and then fine-tune the parameters of the remaining layers of the new model in the target domain.

Additionally, the specific TL methods of the proposed strategies are highly dependent on the structure of the adopted DL model, particularly the depth of a network which is reflected in the layer number of the DL model. Considering the complexity of the LSTM unit and the volume of the adopted dataset, the number of LSTM layers within a deep neural network is set as one in this work (normally set as one to three). As a result, the DL model with a single-layer LSTM combined with a linear layer in series is adopted to verify the feasibility and effectiveness of the proposed TL strategies. According to the definitions of the proposed TL strategies, six specific TL methods for the DL model with explanations are summarized in Table 1.

It is noted that different TL methods may exhibit a variety of performances and evaluations of the above TL methods are provided in Subsection 5.2. In the next section, a transferrable framework, i.e., the constrained LSTM model, is given in detail.

3. Transferable framework: constrained long short-term memory model

For the standard LSTM, it may produce physically unreasonable predictions during training and forecasting processes [24–26], without considering some prior knowledge. Therefore, the method of acquiring physical constraints from prior knowledge via the K-nearest neighbors (KNN) algorithm is introduced in Subsection 3.1. Subsequently, a constrained long short-term memory model (C-LSTM) is developed based on the standard LSTM by integrating the acquired constraints in Subsection 3.2.

3.1. Acquisition of physical constraints

The main idea of KNN is that, whenever there is a new point to predict, its K nearest neighbors are selected from the training dataset [35]. In this work, KNN is used to find the appropriate historical samples with characteristics similar to the future weather condition provided by the numerical weather prediction (NWP) model. Then, incorporating the LSTM model, the PV power observations of these historical samples will be extracted and used to construct the bound conditions to constrain the training process of the C-LSTM model. The main process of the C-LSTM model based on KNN can be expressed simply as the following four steps:

Step 1 Calculate the pre-defined distance between the objective sample and the training samples in sequence.

Step 2 Select K nearest neighbors as similar days from the training samples with the K smallest distances.

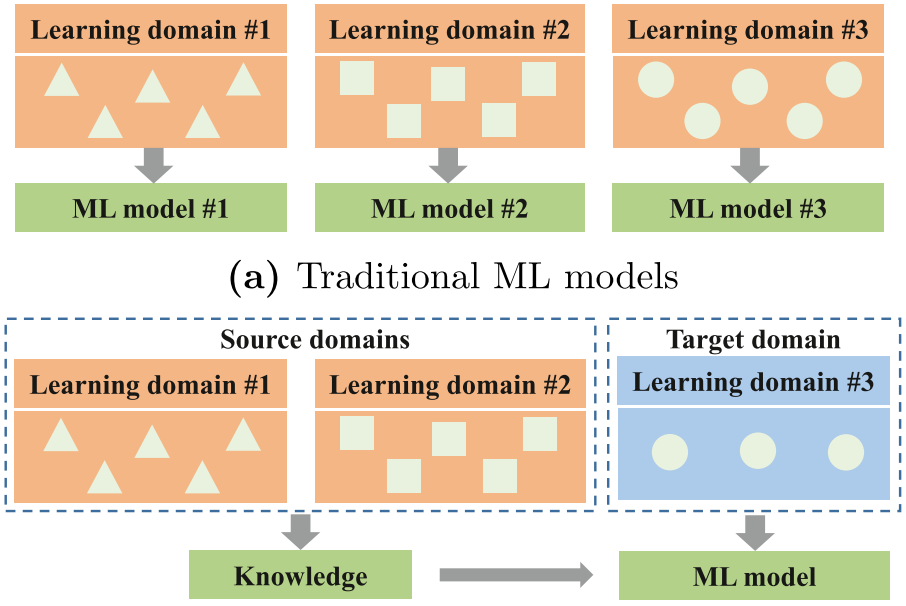


Fig. 1. Different learning processes between models.

Step 3 Determine the bound conditions from the observations of K nearest samples, and verify the effectiveness of the constructed conditions.

Step 4 Integrate bound conditions as physical constraints into the DL model to guide the training process of the model.

Firstly, a distance measure is required to assess the similarity of any two instances. In practice, the Euclidean distance, the Mahalanobis distance, and the Manhattan distance [35] are the most commonly used distance metrics. In this work, the Euclidean distance metric is adopted and enhanced by using weighting as follows.

$$D(X^{t}, X^{c}) = \sqrt{\sum_{i=1}^{n_v} w_i \cdot (x_i^{t} - x_i^{c})^2} \quad (1)$$

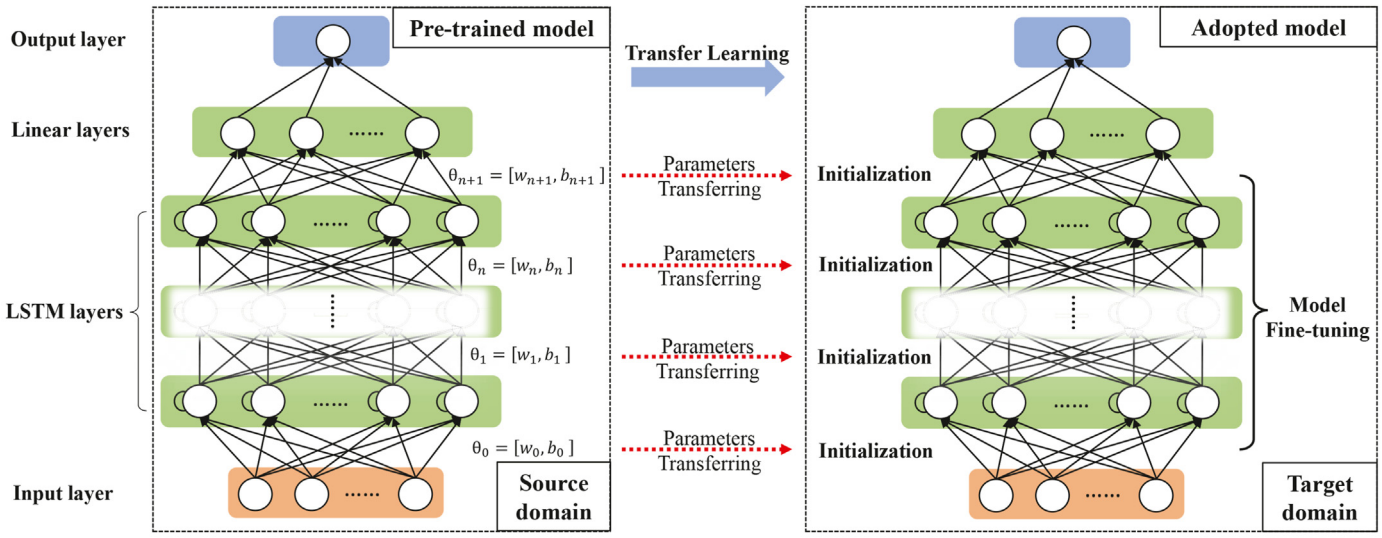
where X^{t} and X^{c} are the objective instance and the compared instance from the training dataset, respectively; x_i^{t} and x_i^{c} are input variables (e.g., meteorological variables, such as solar radiation, temperature, and humidity) in X^{t} and X^{c}, respectively; n_v represents the number of input variables; and w_i is the weight assigned to the ith input variable. In the original Euclidean distance, the weight w_i is equal to one, which means that each input variable is assumed to have the same contribution to the distance D. Nonetheless, the importance of each input variable to the PV power output is apparently different. For example, the solar radiation provided by the NWP model is the most relevant and significant input variable for the determination of the final PV output. As a result, it should be equipped with a higher value of weight compared to other input variables. In the weighted Euclidean distance, the proposed weight w_i reflects the contribution made by each input variable to the final distance. In this work, the weight w_i is determined according to the relevance between the input variable and the target variable by using the Pearson correlation coefficient (PCC) [5,36]. By doing so, the input variable which has a higher correlation with the target variable can have a higher value of weight in distance calculation. It is noted that the absolute value of the calculated PCC is assigned to the corresponding weight w_i in this work.

Secondly, the instances with the K smallest distances are chosen as the K nearest neighbors. Let $X^{nb} = [X_1^{nb}, X_2^{nb}, \ldots, X_K^{nb}]$ be the K nearest neighbors, and their corresponding observations of PVPG are denoted as $Y^{nb} = [y_1^{nb}, y_2^{nb}, \ldots, y_K^{nb}]$. The input variables of the objective instance X^{obj} are very similar to its K nearest neighbors X^{nb}. Therefore, the output of the objective instance, y, should have a high probability to fall into a determined range, $R = [y_{min}^{nb}, y_{max}^{nb}]$, where R is strongly dependent on the selection of K value. Now, given an initial hypothesis K, the problem is transformed into assessing the hypothesis to determine if it can enable objective observations to fall into the determined ranges with a satisfied probability p, where $p = \text{prob}(y \in R|K)$.

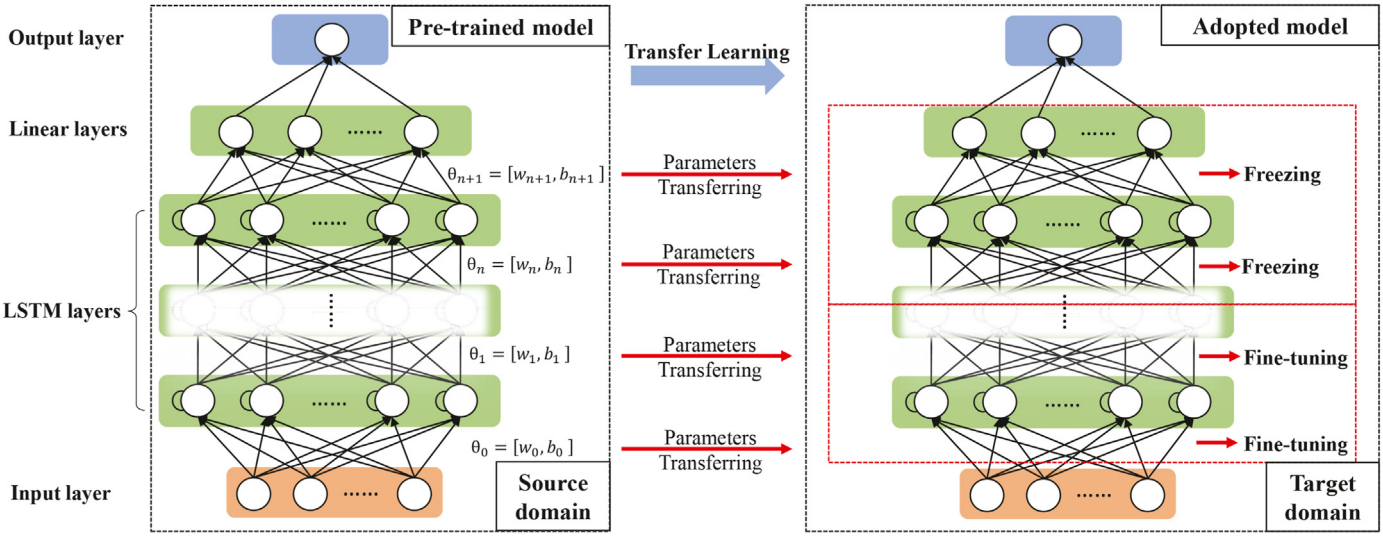
In the K value selection process, the requirement of an acceptable hypothesis of K can be defined as: if $p < 1 - \alpha$, the hypothesis of the current K is rejected, and we tend to have $K + 1$ neighbors. The evaluation process continues with the increase of the values of K until it satisfies the condition $p \geqslant 1 - \alpha$, where α is a preset significance level (SL) and generally set to 0.01 or 0.05 in statistical hypothesis [36,37]. This indicates that the probability of the objective observation falling into the constructed boundaries is equal to 99% ($\alpha = 0.01$) or 95% ($\alpha = 0.05$). In this way, the extracted bound conditions by KNN can be theoretically proven to be valid. The results of K value determination for two target domains in NCPV plants are proposed in Subsection 5.2.

3.2. Proposed constrained long short-term memory model

The C-LSTM model is developed based on the standard LSTM model by properly integrating physical constraints. As a consequence, the bound conditions extracted from prior knowledge by KNN in the previous subsection are integrated into the construction of the C-LSTM model. The architecture of the proposed C-LSTM model is presented in Fig. 3. The integration of these constraints into C-LSTM is achieved by reconstructing the loss function. Theoretically, when the bound conditions are violated, there should be a penalty loss term or knowledge-based loss term, MSE_{PLT},



(a) Strategy #1 - shallow transferring



(b) Strategy #2 - deep transferring

Fig. 2. Proposed parameter-transferring TL strategies.

Table 1
Illustrations of specific methods of the TL strategies for the adopted model structure.

TL strategy	TL methods	Method explanation
Strategy #1	C.1.2	Transfer parameters in Layer #1 and #2 of pre-trained model in source domain
	C.1	Transfer parameters in Layer #1 of pre-trained model in source domain
	C.2	Transfer parameters in Layer #2 of pre-trained model in source domain
Strategy #2	F.0	Fine-tune parameters in all layers without freezing any layers in target domain
	F.1	Freeze parameters in Layer #1 and fine-tune the remaining layers in target domain
	F.2	Freeze parameters in Layer #2 and fine-tune the remaining layers in target domain

reflected in the loss function [38]. Consequently, the loss function of the C-LSTM can be reformulated as follows.

$$\mathcal{L}(\theta)_{\mathrm{C-LSTM}} = \mathrm{MSE}_{\mathrm{DATA}} + \mathrm{MSE}_{\mathrm{PLT}} \tag{2}$$

$$\mathrm{MSE}_{\mathrm{DATA}} = \frac{1}{N_{\mathrm{DATA}}} \sum_{i=1}^{N_{\mathrm{DATA}}} |\hat{y}_i - y_i|^2 \tag{3}$$

$$\mathrm{MSE}_{\mathrm{PLT}} = \frac{1}{N_{\mathrm{PLT}}} \sum_{i=1}^{N_{\mathrm{PLT}}} \lambda_{\mathrm{PLT},i} \cdot |\hat{y}_i - y_i^*|^2 \tag{4}$$

where

The total loss of C-LSTM is reconstructed as the combination of data loss and penalty loss, as illustrated in Eq. (2). In Eq. (3), $\hat{y}_i$ and y_i are the ith forecast and observation, respectively. In Eq. (4), $y_i^* = \frac{1}{2}\left(y_{\min,i}^{\mathrm{nb}} + y_{\max,i}^{\mathrm{nb}}\right)$; and $\lambda_{\mathrm{PLT},i}$ is an additional hyper-parameter of C-LSTM, which denotes the intensity of penalty on the loss function. During the training process, when an output after the linear layer in the network, $\hat{y}_i$, satisfies the bound conditions, $\lambda_{\mathrm{PLT},i} = 0$ and $\mathrm{MSE}_{\mathrm{PLT}} = 0$, this means that no penalty is added to the loss function and the model is trained as it was previously. In contrast, when the bound conditions are violated, $\lambda_{\mathrm{PLT},i} > 0$ and $\mathrm{MSE}_{\mathrm{PLT}} > 0$, this indicates that additional penalties will be added to the loss function of C-LSTM. Meanwhile, the parameter vector, $\theta = \{W, b\}$, will be tuned accordingly via an optimizer, such as adaptive moment estimation (Adam) or stochastic gradient descent (SGD). Therefore, $\lambda_{\mathrm{PLT},i}$ can be illustrated as follows.

$$\begin{cases} \lambda_{\mathrm{PLT},i} = 0, & \hat{y}_i \in [y_{\min,i}^{\mathrm{nb}},\ y_{\max,i}^{\mathrm{nb}}] \\ \lambda_{\mathrm{PLT},i} > 0, & \hat{y}_i \notin [y_{\min,i}^{\mathrm{nb}},\ y_{\max,i}^{\mathrm{nb}}] \end{cases} \tag{5}$$

The integrated constraints can assist the training process of the C-LSTM model and reduce physically unreasonable predictions that may occur in the conventional LSTM model, thus improving the model performance. Compared to the methods in Ref. [5], the advantage of using KNN to extract physical constraints is that this method is effective not only for data on a large scale, but also for data on a small scale. It is noted that this work relies partially on our previous work in Ref. [5].

4. Data preparation

In Subsection 4.1, the utilized datasets in this work are described in detail. Subsequently, the initial datasets are pre-processed in Subsection 4.2. The adopted evaluation metrics are introduced at last in Subsection 4.3.

4.1. Data description

Real-life PV datasets from two individual PV plants located in different areas of Australia are utilized. Meanwhile, feature datasets containing weather forecasts for eight commonly-used weather variables are obtained from the European Center for Medium-range Weather Forecasts (ECMWF) [39]. Installation parameters of the adopted PV plants and explanations of input weather variables are introduced in Table 2 and Table 3, respectively.

Table 2
Installation parameters of adopted PV plants [39].

Parameter	Plant #1	Plant #2
Altitude	595 m	602 m
Panel type	Solarfun SF160-24-1M195	Suntech STP190S-24/Ad+
Panel number	8	26
Nominal power	1560 W	4940 W
Orientation	38°, Clockwise from North	327°, Clockwise from North
Panel tilt	35°	35°

In addition, to ensure convincible results, datasets from two PV plants are cross-validated. In practice, one dataset with a large volume of data is set as the source domain, and limited data from another one are set as the target domains. As a result, datasets include two source domains and several target domains are established, and simultaneously used for method verifications. In this study, the source domain contains data of 27 months, whereas the target domain only includes data of 40 days. In addition, the forecasting horizon depends entirely on the horizon of weather forecasts that can be obtained in advance. It is assumed that accurate one day ahead weather forecasts are available. Therefore, the PVPG forecasting horizon can be regarded as day-ahead. It is noted that the proposed methods can also produce PVPG forecasts in a shorter time interval (e.g., 15 min, 5 min, and 1 s). However, the forecasting interval is entirely dependent on the time interval of available data. In this work, the data (i.e., weather data and PV data) obtained from the ECMWF and the adopted PV plants both are collected hourly. Hence, we focus on the research of hourly PVPG forecasting.

Moreover, to evaluate the correlations between the target variable and the feature variables listed in Table 3, the PCC criterion [5,36] is utilized. The evaluation results illustrated in Fig. 4 demonstrate that the feature variables SR and TS, both related to solar radiation, have very strong correlations with PVPG, which means the output of a PV system is highly dependent on the solar-radiation-related variables. The obtained results are also consistent with the previous discussion.

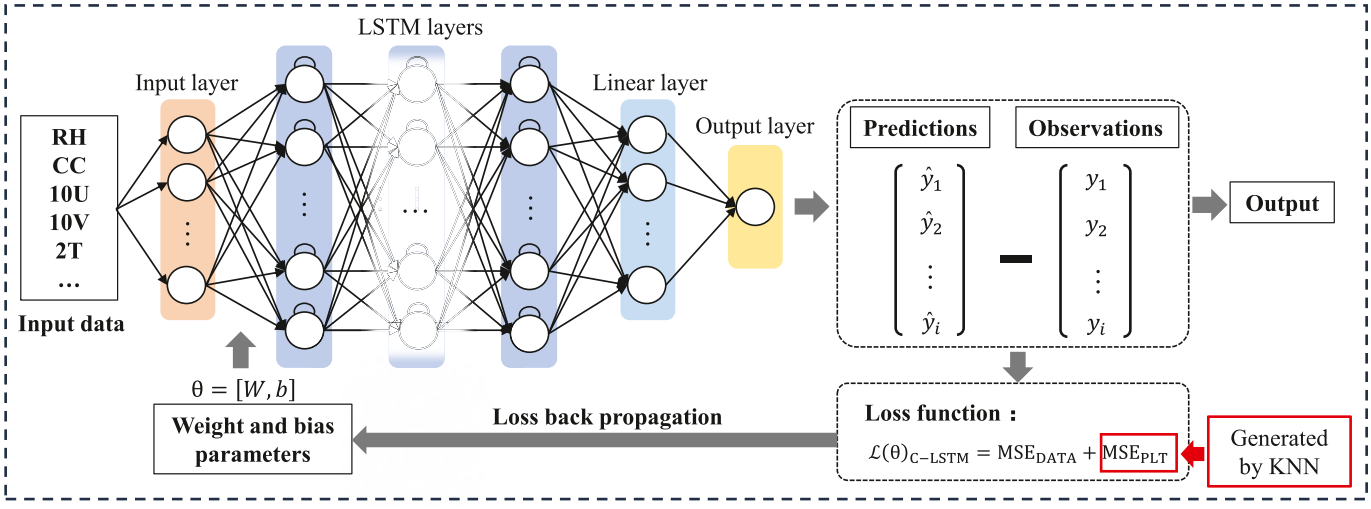


Fig. 3. Architecture of the C-LSTM model by using KNN to extract constraints.

Table 3
Illustration of available feature variables within PV datasets [39].

ID	Feature name	Unit	Comment(s)
RH	Relative humidity at 1000 mbar	%	Defined with respect to saturation of the mix phase.
CC	Total cloud cover	0–1	Derived using the model's overlap assumption.
10U	10-m U wind component	$m \cdot s^{-1}$	Wind speed at 10-m along the longitudinal direction.
10V	10-m V wind component	$m \cdot s^{-1}$	Wind speed at 10-m along the latitudinal direction.
2T	2-m temperature	K	Temperature at 2-m.
SR	Surface solar radiation downwards	$J \cdot m^{-2}$	Solar radiation at the land surface.
TR	Surface thermal radiation downwards	$J \cdot m^{-2}$	Thermal radiation at the land surface.
TS	Top-net solar radiation	$J \cdot m^{-2}$	Net solar radiation at the top of the atmosphere.

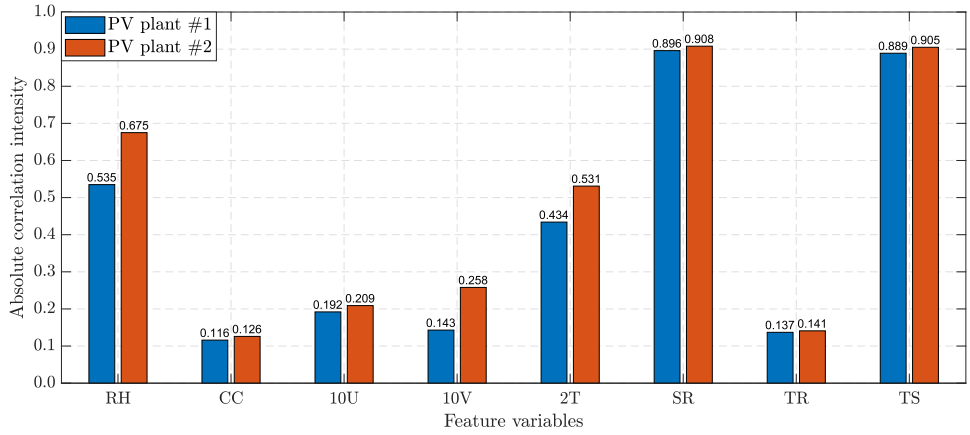


Fig. 4. Evaluation of correlations between the target variable and the feature variables.

4.2. Data pre-processing

4.2.1. Data normalization

Data normalization aims to process data entries in various ranges into a consistent range, thus minimizing regression error while maintaining correlation among the dataset. The min-max normalization, which restricts the data within the range between zero and one, is adopted in this work. The formula of min-max normalization can be expressed as:

$$x_{\text{norm}} = \frac{x_{\text{orig}} - x_{\min}}{x_{\max} - x_{\min}} \tag{6}$$

where x_{norm} and x_{orig} are the normalized data and original data, respectively; and $x_{\min}$ and $x_{\max}$ denote the minimum and maximum values of a specific feature within the dataset, respectively. All feature data should be normalized before transmitting to DL models.

4.2.2. Data splitting for cross-validation

Cross-validation is an effective way to ensure that forecasting models are robust. In cross-validation, the initial dataset is usually divided into two sets, i.e., the training dataset and the testing dataset. The former dataset with the bulk of data is used for training the model, while the latter dataset is used to evaluate the model to prevent over-fitting. In the source domain, the historical data of 730 days (24 months) are set as the training dataset, whereas the data of 91 days (3 months) are set as the testing dataset. In addition, in the target domain, the historical data of 30 days are set as the training dataset, whereas the data of 10 days are set as the testing dataset. The method of cross-validation ensures that any improvements achieved by the proposed model based on the training data will also lead to an improvement for the testing data, thus enhancing the robustness of the model.

4.3. Evaluation metrics

To assess the forecasting performances of models from different perspectives, except for mean square error (MSE), we employ other three popularly used criteria, i.e., root mean square error (RMSE) [40], normalized root mean square error (NRMSE) [41], and the forecasting skill [42], denoted as S, which measures the improvement of the proposed forecasting method over the reference persistence model, as the evaluation metrics. Formulations of MSE, RMSE, NRMSE, and the forecasting skill can be mathematically illustrated in Eqs. (7) - (10), respectively.

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N} |\hat{y}_i - y_i|^2 \tag{7}$$

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} |\hat{y}_i - y_i|^2} \tag{8}$$

$$\text{NRMSE} = \frac{\text{RMSE}}{\sum_{i=1}^{N} y_i} \tag{9}$$

$$S = 1 - \frac{\text{RMSE}}{\text{RMSE}_{\text{p}}} \tag{10}$$

where RMSE_{p} is the RMSE of the reference persistence model; symbols $\hat{y}_i$, y_i, and $\bar{y}$ represent the forecasted value, the observed value, and the mean of observed values, respectively. It is important to note that the data of PVPG have already been normalized within the initial datasets. Hence, the unit of MSE, RMSE, and NRMSE can be regarded as 1.

Table 4
Illustration of adopted hyper-parameters within pre-trained models.

Source domain	Hyper-parameters					
	LSTM layer number	Time step	Hidden unit	Input size	Learning rate	Batch size
Source domain #1 (dataset from plant #1)	1	4	16	8	0.001	256
Source domain #2 (dataset from plant #2)	1	4	16	8	0.005	512

5. Case study and result

In Section 5, several case studies are carried out to evaluate the feasibility and effectiveness of the proposed models.

5.1. Model pre-training in source domain

A standard LSTM model has to be pre-trained in the source domain at first and then transferred to the target domain for fine-tuning via the proposed TL strategies. The pre-training process assists models to learn general knowledge that can be shared between different but related tasks. As discussed in Subsection 4.1, datasets from two independent PV plants are simultaneously utilized as the source domains. Hence, two pre-trained LSTM models are prepared for the following evaluations.

In DL, the setting of hyper-parameters of the model may cause significant impacts on model performance. After training the LSTM models in the source domains multiple times, the suitable hyper-parameters of models are determined as in Table 4. The initial network parameters are set by default, and the Adam algorithm is set as the optimizer in the pre-training processes. The models are trained to minimize the sum of MSE of forecast errors. As a result, the variations of training and testing errors for the pre-trained LSTM models during 300 epochs are shown in Fig. 5. The model with the lowest testing errors will be saved and transferred to target domains via different TL methods.

5.2. Evaluation of proposed transfer learning strategies combined with C-LSTM

In this subsection, combined with the transferable framework, i.e., the proposed C-LSTM model, the forecasting performances of different TL strategies are evaluated and the most appropriate TL methods are determined accordingly. If the proposed methods have consistent performances in both target domains (defined in Subsection 4.1), the methods can be verified.

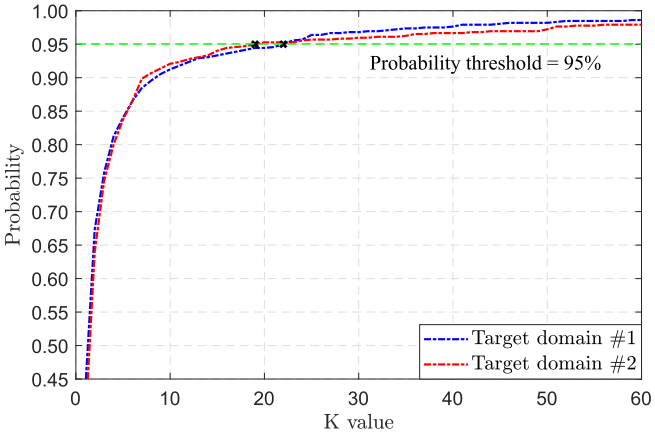


Fig. 6. Probability variations with K value increasing. Significance level $\alpha = 0.05$.

One prerequisite task prior to PVPG forecasting by C-LSTM is to determine an appropriate K value. Based on the determination process of the K value illustrated in Subsection 3.1, the significance level α is set as 0.05 in this work. This indicates that the probability of the objective observation falling into the constructed boundaries is over 95%. Subsequently, the probability variations with K value increasing for both target domains are presented in Fig. 6. Consequently, $K_{TD1} = 22$ and $K_{TD2} = 19$ are utilized while using KNN to construct bound conditions as physical constraints in C-LSTM.

In addition, each experiment has to be conducted multiple times ($N_{test} = 10$) to reduce the impact of randomness in all comparisons. The same hyper-parameters are adopted for different TL methods to ensure the fairness of comparisons. Subsequently, the evaluation

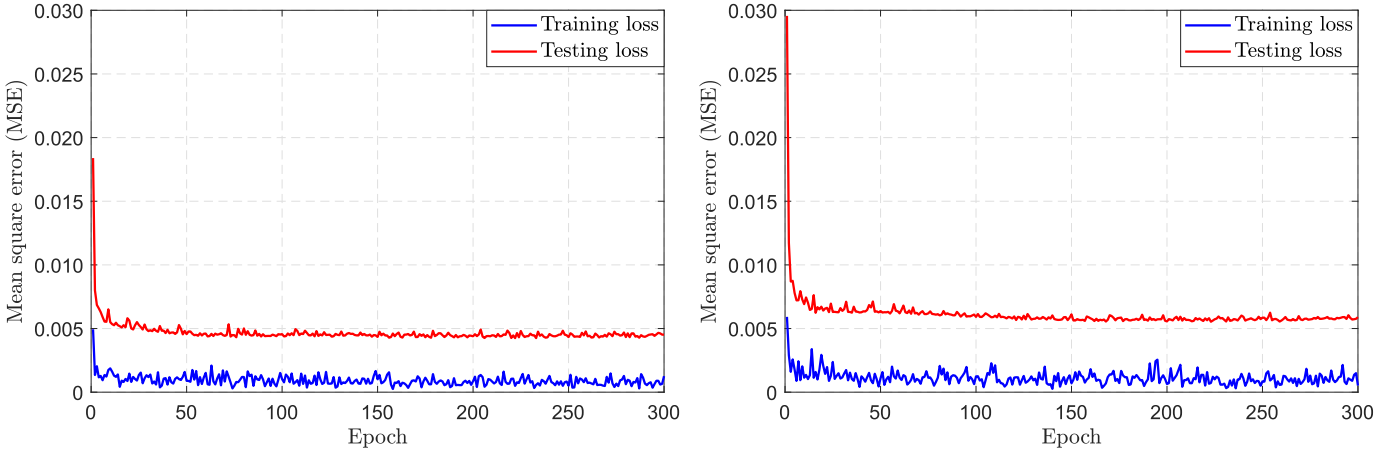


Fig. 5. Variations of training and testing errors for the pre-trained LSTM models.

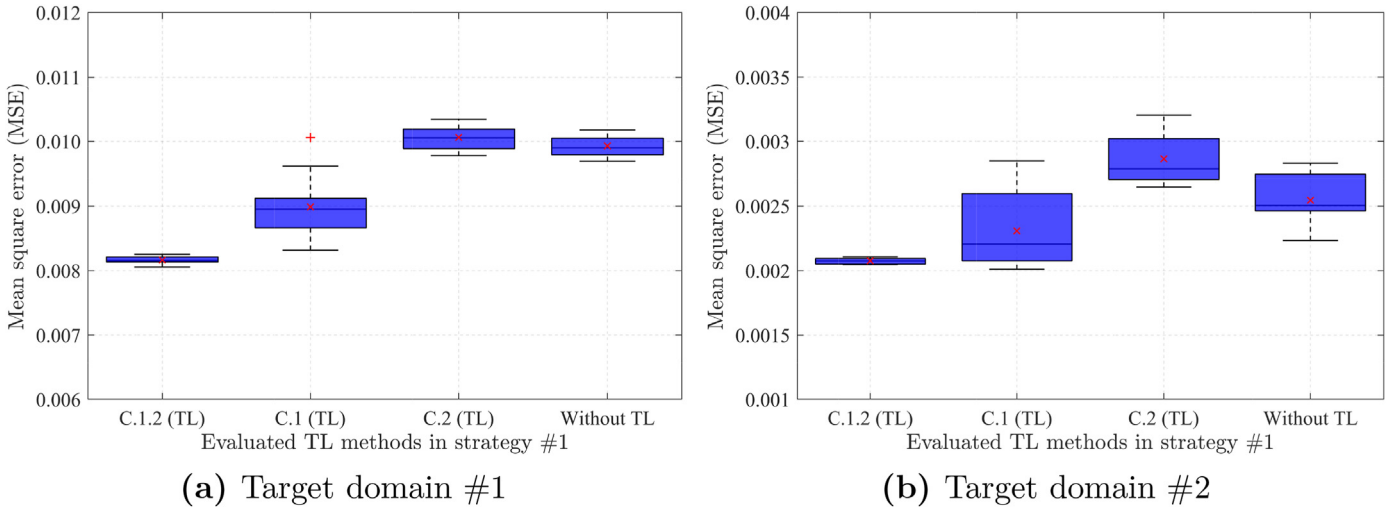


Fig. 7. Box-plot results based on C-LSTM assisted by TL strategy #1.

results for the adopted DL structure (i.e., single-layer LSTM combined with a linear layer in series as discussed in Subsection 2.2) are illustrated in Fig. 7 - Fig. 8, respectively. In the figures, the blue boxes represent the MSE loss of C-LSTM assisted by different TL methods. The red crosses denote the averaged MSE for multiple evaluations, whereas the red pluses are outliers accidently produced by models. The standard deviation (SD), which can be used to describe the stability of the model, is reflected on the bound of boxes. Overall, the tighter is the bound, the more stable is the model.

First, Fig. 7 illustrates the evaluation results based on C-LSTM assisted by TL strategy #1 for two target domains. From the results in both Fig. 7 (a) and (b), it can be seen that properly using TL methods of strategy #1, such as C.1.2 (or C.1), can considerably improve PVPG forecasting accuracy by 18.79% and 17.20% for target domain #1 and target domain #2, respectively, when compared to C-LSTM without TL. This is because the transferable knowledge, which reflects the common features, is usually stored in the shallow layers (i.e., the LSTM layers). Transferring parameters of the shallow layers of the pre-trained model to the model in the target domain as initialization by C.1.2 (or C.1) can help to improve the forecasting ability of the model and result in better performance. Additionally, it is interesting to observe that independently transferring parameters of the deep layer (i.e., the linear layer) to the model in the target domain by C.2 may have negative impacts on performance improvement. The linear layer (also called the output layer) of a DL network is only compatible with the corresponding hidden layers. Hence, it is not useful to transfer the parameters of the output layer separately.

Subsequently, the evaluation results based on C-LSTM assisted by TL strategy #2 for two target domains are shown in Fig. 8. According to the results, it is apparent that properly using TL methods of strategy #2, such as F.2 (or F.0), can also significantly improve forecasting accuracy by 19.63% and 24.77% for target domain #1 and target domain #2, respectively, when compared to C-LSTM without TL. Besides, compared with the results of strategy #1 in Fig. 7, it is obvious that the C-LSTM model assisted by TL strategy #2 outperforms the model assisted by strategy #1 with superior stability (tighter bound of the box). Since some parameters of layers are fixed by using strategy #2, the range of model output is restricted to some extent.

The numerical results of different TL methods combined with C-LSTM are presented in Table 5. The results are expressed as the averaged MSE plus or minus SD of multiple evaluations. The lower

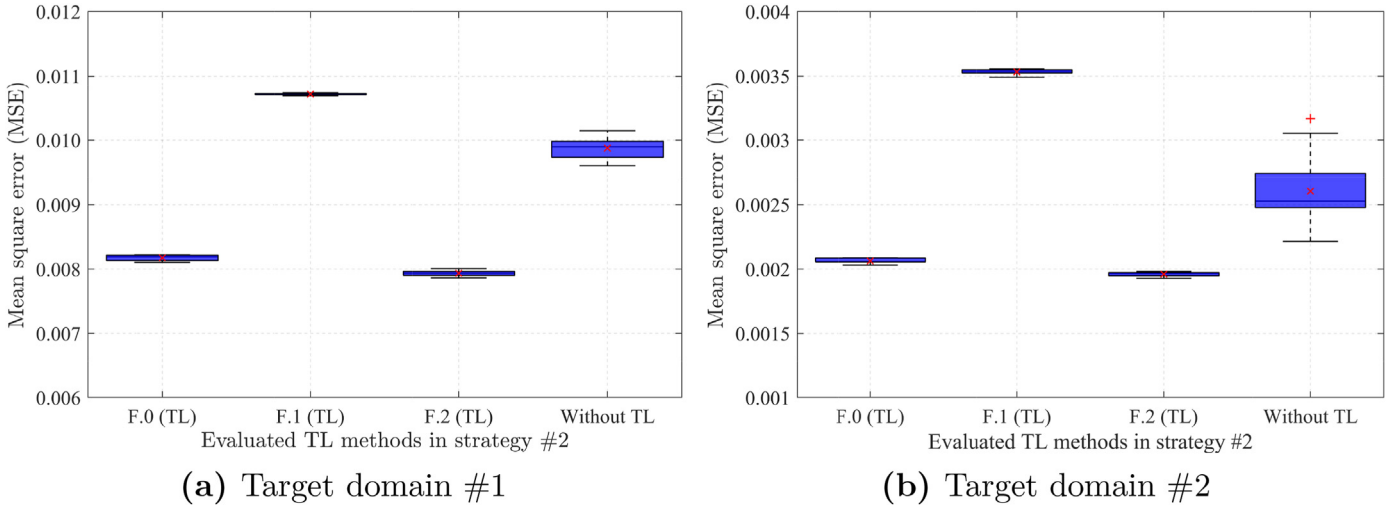


Fig. 8. Box-plot results based on C-LSTM assisted by TL strategy #2.

Table 5
Numerical results of different TL methods combined with C-LSTM.

TL strategy	TL methods	Target domain #1		Target domain #2	
		MSE$^{\times 10^{-3}}$	Imp.rate	MSE$^{\times 10^{-3}}$	Imp.rate
Strategy #1	C.1.2	***8.163 ± 0.060***	***18.79%***	***2.076 ± 0.023***	***18.42%***
	C.1	8.987 ± 0.530	9.49%	2.308 ± 0.285	9.31%
	C.2	10.060 ± 0.192	−1.32%	2.865 ± 0.187	−12.58%
	Without TL	9.929 ± 0.158	–	2.544 ± 0.205	–
Strategy #2	F.0	8.177 ± 0.042	17.20%	2.064 ± 0.019	20.80%
	F.1	10.713 ± 0.013	−8.48%	3.533 ± 0.020	−35.60%
	F.2	***7.937 ± 0.047***	***19.63%***	***1.960 ± 0.018***	***24.77%***
	Without TL	9.875 ± 0.161	–	2.605 ± 0.304	–

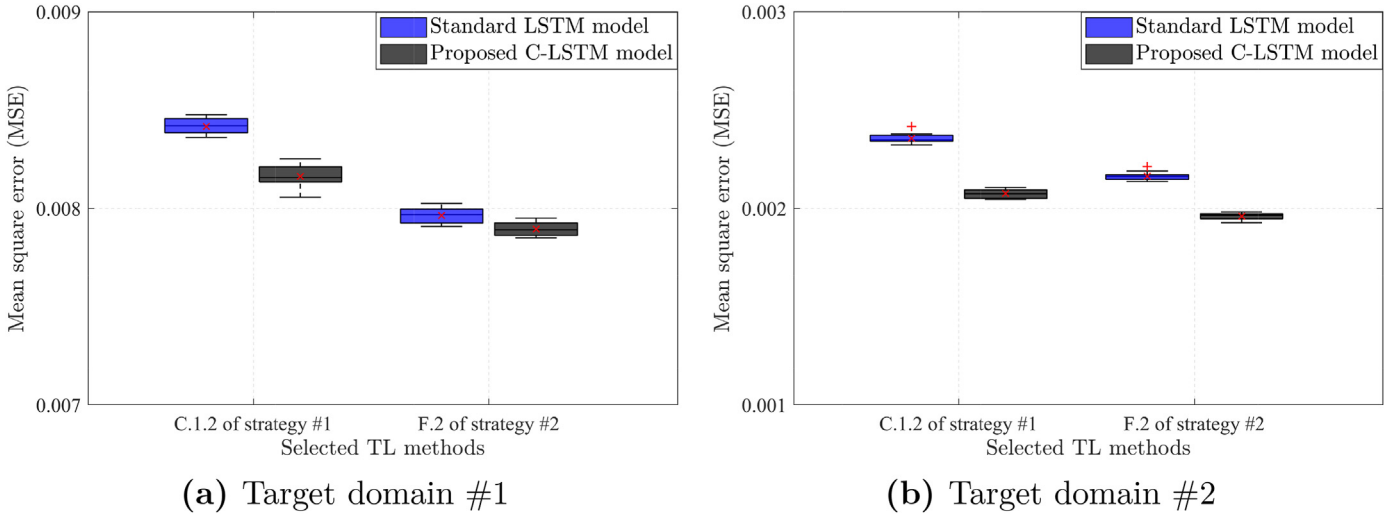


Fig. 9. Comparisons between LSTM and C-LSTM assisted by the selected TL methods.

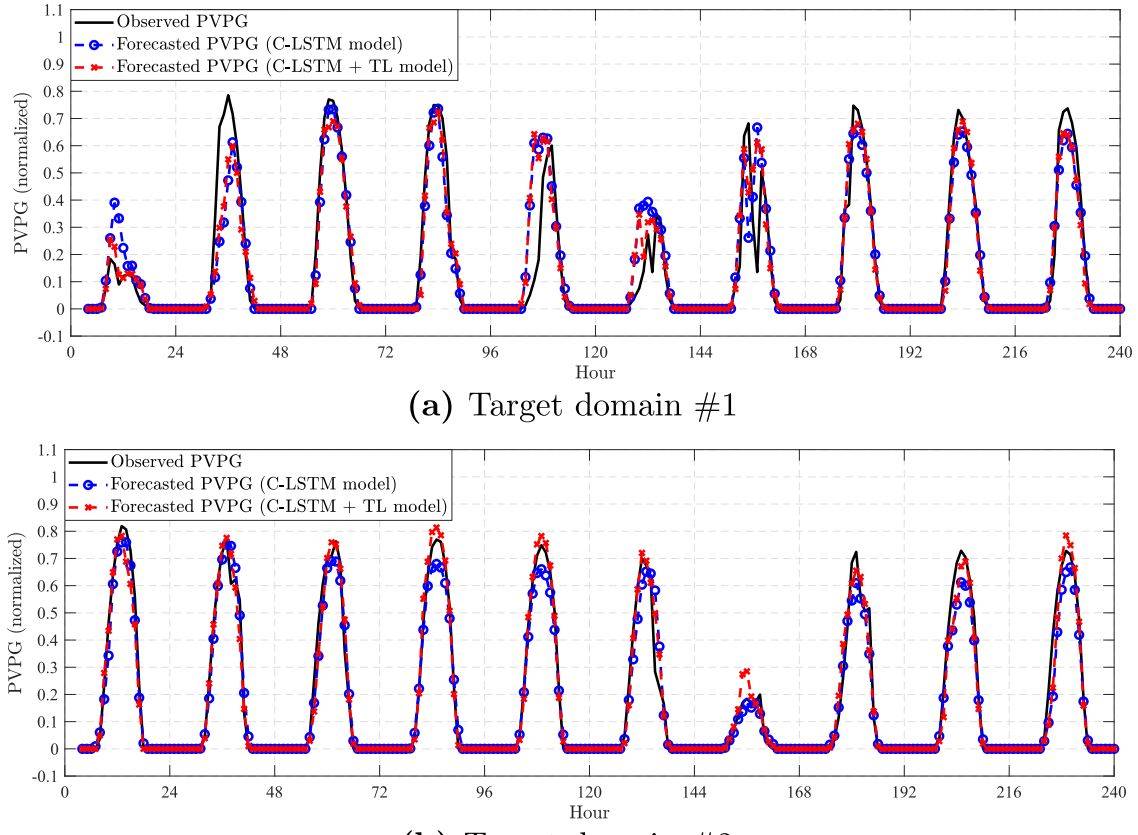


Fig. 10. Examples of visually displayed PVPG forecasting results based on proposed models.

values represent greater accuracy and stability of the model. Additionally, the improved rate of proposed TL methods compared to the method without using any TL techniques is also presented to verify the proposed approaches. The best results are marked in bold. In summary, the results indicate that the method of C.1.2 in strategy #1 and the method of F.2 in strategy #2 are the most effective ones among all TL methods. Therefore, C.1.2 and F.2 are determined as the appropriate TL methods for PVPG forecasting in this work. However, not all proposed methods are effective for promoting the accuracy of PVPG forecasting. In some cases, the performances of DL models will be considerably degraded by using inappropriate transfer methods.

5.3. Comparison between LSTM and C-LSTM incorporated with determined TL methods

Since the proposed C-LSTM model is developed based on the standard LSTM model by integrating physical constraints via KNN, it is necessary to compare the forecasting performance of C-LSTM versus LSTM. Based on the obtained results in Subsection 5.2, the TL methods, i.e., C.1.2 and F.2 which performed the best in strategy #1 and strategy #2, respectively, are selected to assist DL models in the following evaluations. The box-plot results of forecasting performance comparison between LSTM and C-LSTM assisted by the selected TL methods are illustrated in Fig. 9.

Specifically, the blue boxes and dark grey boxes in Fig. 9 denote the MSE loss of LSTM and C-LSTM, respectively. From the results, it is obvious that the proposed C-LSTM model outperforms the standard LSTM model with higher accuracy (lower averaged MSE) in all cases for both target domains, due to the contributions made by the integrated constraints which can guide the training process of DL models. Therefore, the obtained results indicate that the proposed C-LSTM model assisted by the selected TL methods is effective for the PVPG forecasting problem of NCPV plants, and also demonstrates superior performance with higher accuracy of PVPG forecasting compared to the standard LSTM model to some extent.

5.4. Day-ahead forecasting of PV power generation

The performances of hourly day-ahead PVPG forecasting for the following ten days in target domain #1 and target domain #2, by different models are compared in Subsection 5.4. In this work, the proposed C-LSTM model assisted by the best TL method (i.e., F.2 of strategy #2, denoted as C-LSTM + TL), is mainly compared with a commonly-used persistence model (i.e., a forecasting model in which the observations of the last day are persisted forward as the forecasting results) [5,42], a statistical model (i.e., the auto-regressive integrated moving average model - ARIMA) [9], conventional machine learning models (i.e., the k-nearest neighbors - KNN [12] and the fully-connected neural network - FCNN [14]), the standard LSTM model [24], the hybrid model (i.e., CNN-LSTM) [17], and the C-LSTM model without TL. It is important to note that the same hyper-parameters and network structures are adopted for LSTM, C-LSTM, and C-LSTM + TL. To prevent the impact of randomness, all experiments are conducted multiple times as usual.

To visually display the forecasting results, examples of PVPG forecasting results based on proposed DL models are presented in Fig. 10. From the results, both compared models can accomplish the task of PVPG forecasting with acceptable accuracies. In general, the forecasted PVPG curves follow the variations of the observed PVPG. Additionally, the forecasted values produced by the C-LSTM + TL model are closer to the observed values than the values produced by C-LSTM, which verifies the feasibility and effectiveness of proposed TL methods.

Subsequently, the numerical results of PVPG forecasting comparisons between different forecasting models are illustrated in Table 6, where the persistence model and the standard LSTM model are set as the reference models to calculate the forecasting skills.

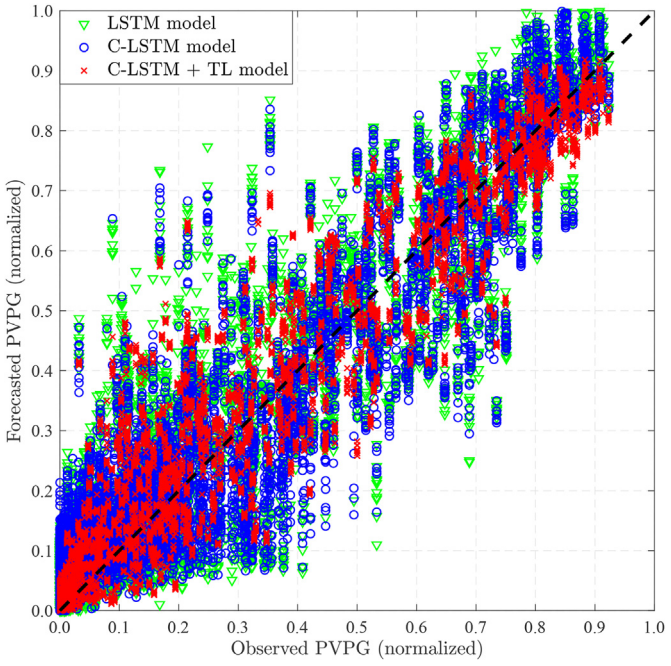
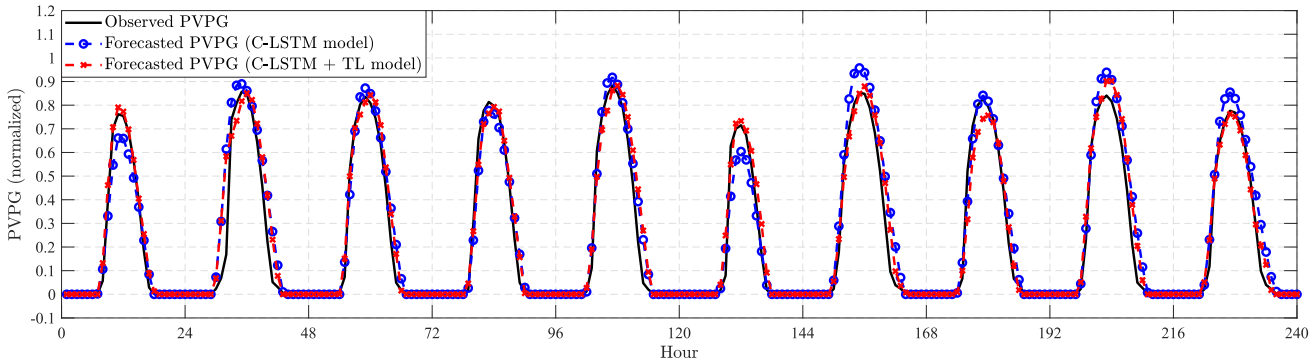


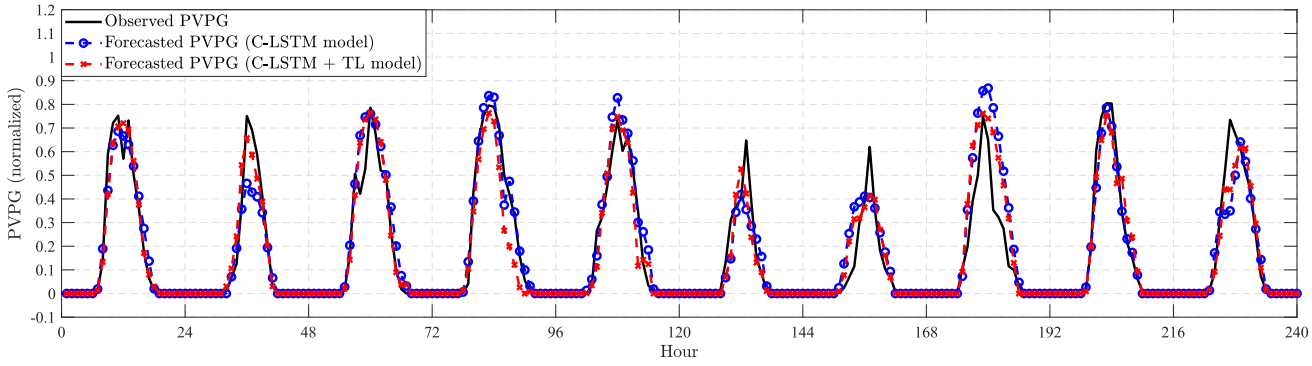
Fig. 11. Scatter plots of observed and forecasted day-ahead PVPG using standard LSTM, C-LSTM, and C-LSTM + TL models (evaluated based on target domain #3 - target domain #8).

Table 6
Numerical results of PVPG forecasting for the following ten days by different forecasting models. (Symbol "-" indicates no improvement compared to the reference model.)

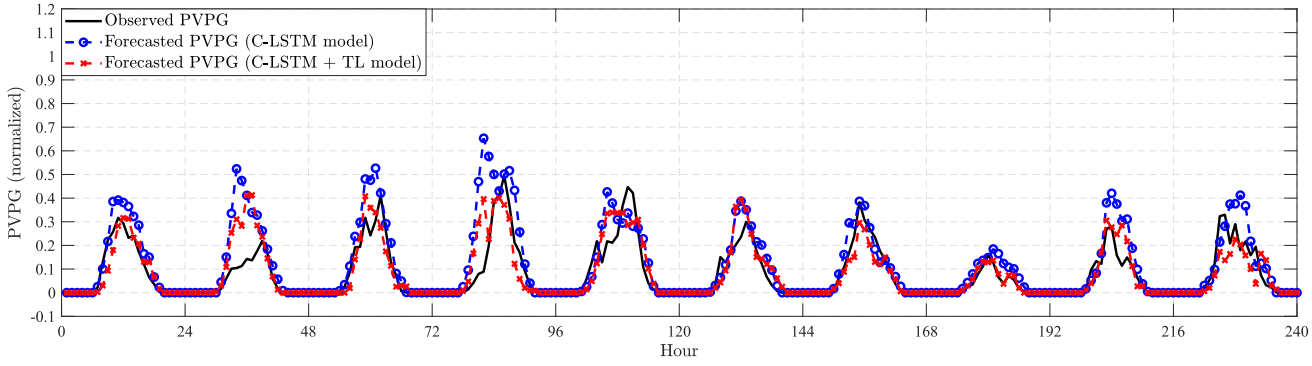
Datasets	Criteria	Forecasting models							
		Persistence	ARIMA	KNN	FCNN	LSTM	CNN-LSTM	C-LSTM	TL + C-LSTM
Target domain #1	$MSE^{\times 10^{-3}}$	24.98	11.47	9.25	11.02 ± 0.97	10.03 ± 0.29	9.56 ± 0.23	9.95 ± 0.25	***7.90 + 0.03***
	$RMSE^{\times 10^{-2}}$	15.81	10.71	9.62	10.49 ± 0.47	10.01 ± 0.14	9.78 ± 0.12	9.98 ± 0.12	***8.89 ± 0.02***
	$NRMSE^{\times 10^{-3}}$	4.55	3.08	2.77	3.02 ± 0.14	2.88 ± 0.04	2.813 ± 0.04	2.87 ± 0.04	***2.56 ± 0.01***
	$S_{persistence}$	–	32.25%	39.14%	33.63%	36.65%	38.15%	36.88%	***43.78%***
	S_{LSTM}	–	–	3.93%	–	–	2.36%	0.37%	***11.26%***
Target domain #2	$MSE^{\times 10^{-3}}$	19.64	11.90	6.07	5.44 ± 0.58	2.64 ± 0.17	2.64 ± 0.22	2.54 ± 0.21	***1.96 + 0.02***
	$RMSE^{\times 10^{-2}}$	14.01	10.91	7.79	7.37 ± 0.38	5.14 ± 0.17	5.13 ± 0.21	5.04 ± 0.20	***4.43 ± 0.02***
	$NRMSE^{\times 10^{-3}}$	3.24	2.52	1.80	1.70 ± 0.09	1.19 ± 0.04	1.19 ± 0.05	1.17 ± 0.05	***1.02 ± 0.01***
	$S_{persistence}$	–	22.15%	44.41%	47.43%	63.36%	63.37%	64.03%	***68.41%***
	S_{LSTM}	–	–	–	–	–	0.03%	7.90%	***13.79%***



(a) Target domain #3: clear sky.



(b) Target domain #4: partly cloudy sky.



(c) Target domain #5: overcast sky.

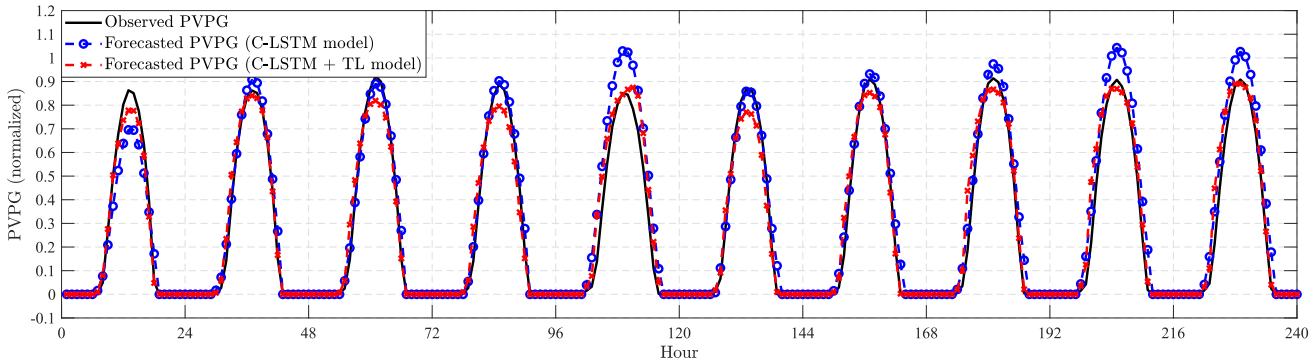
Fig. 12. Examples of visually displayed PVPG forecasting results under different sky conditions. (Evaluated based on datasets from PV plant #1.)

The best results are marked in bold. According to the results, it is apparent that the proposed C-LSTM + TL model performs the best among all compared models with the lowest averaged MSE in both target domains. Specifically, compared to the DL models (i.e., FCNN, LSTM, CNN-LSTM, and C-LSTM), the proposed C-LSTM + TL model can obtain much better PVPG forecasting performance based on a small-size training dataset with lower MSE and lower SD, which demonstrates the superiority of C-LSTM + TL in terms of accuracy and stability. In addition, both LSTM and C-LSTM outperform the conventional machine learning model (i.e., FCNN) with lower MSE in the evaluation, due to the information from previous time steps within the time-series data being used in the training of models. Moreover, the DL models (i.e., LSTM, C-LSTM, and C-LSTM + TL) perform far better than the representative statistical model (i.e., ARIMA) and the persistence model, since the ARIMA model and the persistence model are only developed based on time-series PV data, and meteorological data are not considered. The forecasting skill of C-LSTM + TL can be improved up to 68.4% compared with the persistence model in this study.

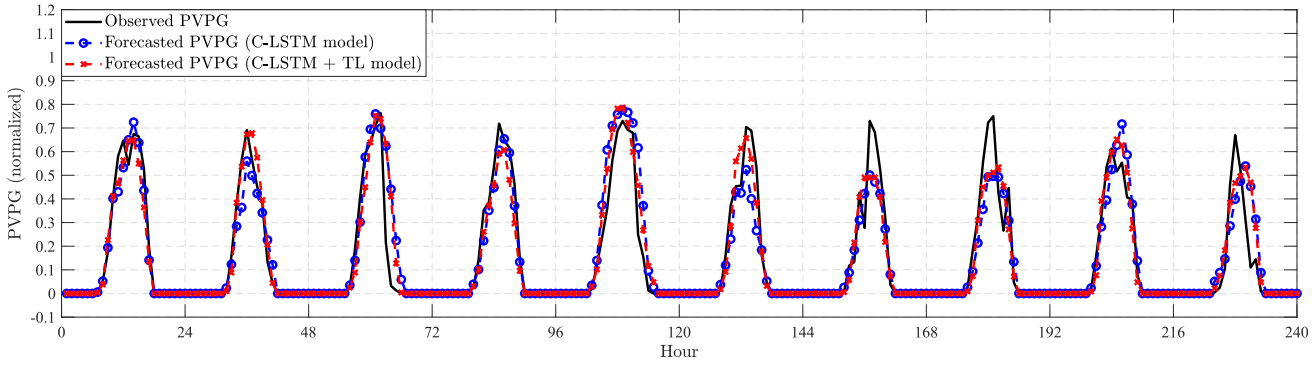
5.5. Forecasting of PV power generation under different sky conditions

To further evaluate the forecasting performances of proposed models under a variety of sky conditions, additional experiments are conducted in Subsection 5.5. According to the total cloud amount of each day (observed from the value of variable CC illustrated in Table 3), testing datasets of two independent PV plants are divided into separate cases for days that are the clear sky, partly cloudy sky, and overcast sky. Therefore, six additional testing datasets (denoted as target domain #3 - target domain #8) are created for further evaluation.

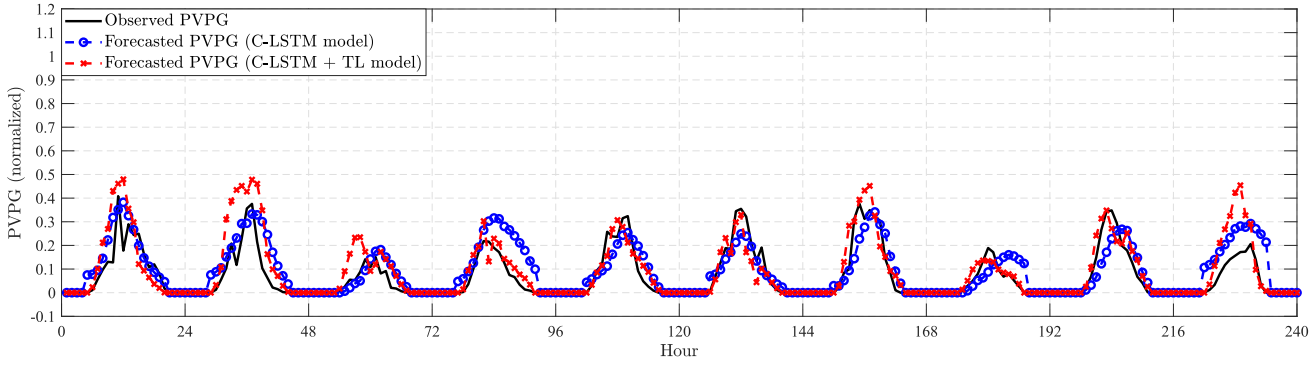
First, scatter plots of observed and forecasted day-ahead PVPG for the above cases using standard LSTM, C-LSTM, and C-LSTM + TL



(a) Target domain #6: clear sky.



(b) Target domain #7: partly cloudy sky.



(c) Target domain #8: overcast sky.

Fig. 13. Examples of visually displayed PVPG forecasting results under different sky conditions. (Evaluated based on datasets from PV plant #2.)

models are presented in Fig. 11. From the obtained results, the scatter plot of C-LSTM + TL shows better forecasting performances than the other two DL models. We can see much lower dispersion between the forecasted and observed PVPG in the overall evaluation by the C-LSTM + TL model. In addition, the newly proposed C-LSTM model performs slightly better than the standard LSTM. This indicates that the use of physical constraints extracted from prior knowledge to guide the training process can boost the forecasting capability of the basic LSTM model with considerable improvement.

Subsequently, the examples of visually displayed PVPG forecasting results under a variety of sky conditions (i.e., clear sky, partly cloudy sky, and overcast sky) by the proposed DL models are specifically presented in Fig. 12 and Fig. 13. The achieved results demonstrate the great forecasting capabilities of the proposed DL models for testing sets of different sky patterns. In general, both C-LSTM and C-LSTM + TL models can capture the major variations of observed PVPG with acceptable accuracies. Compared with the C-LSTM model, the C-LSTM model combined with TL can produce more accurate PVPG forecasts with closer values to the observed PVPG in all cases.

In addition to these, to discuss the results more specifically, the proposed C-LSTM + TL model can astonishingly forecast the PVPG with extremely high accuracies under the clear sky condition, as illustrated in Figs. 12 (a) and 13 (a). However, in the cases of partly cloudy sky and overcast sky, the C-LSTM + TL model could successfully follow the variations of observed PVPG with high accuracies in most days, whereas some detailed variations of PVPG with high variability (e,g., 144th hour - 168th hour in target domain #4, 24th hour - 48th hour in target domain #5, and 168th hour - 192nd hour in target domain #7) are not forecasted accurately by all models. This is because (1) the utilized input weather data are obtained from NWP models, which contain a certain amount of forecasting errors; (2) the training data of NCPV plants are limited,

Table 7
Numerical results of PVPG forecasting under different sky conditions by different models. (Symbol "-" indicates no improvement compared to the reference model.)

Datasets	Criteria	Forecasting models						
		Persistence	KNN	FCNN	LSTM	CNN-LSTM	C-LSTM	TL + C-LSTM
PV Plant #1 Target domain #3 (Clear sky)	$MSE^{*\times10^{-3}}$	12.91	12.75	10.62 ± 2.10	7.57 ± 0.47	6.19 ± 0.65	6.74 ± 0.71	***4.55 ± 0.10***
	$RMSE^{\times10^{-2}}$	11.36	11.29	10.26 ± 1.02	8.70 ± 0.27	7.86 ± 0.40	8.20 ± 0.44	***6.74 ± 0.07***
	$NRMSE^{\times10^{-3}}$	2.13	2.12	1.92 ± 0.19	1.63 ± 0.05	1.47 ± 0.07	1.54 ± 0.08	***1.26 ± 0.01***
	$S_{persistence}$	–	0.61%	9.67%	23.45%	30.80%	27.83%	***40.64%***
	S_{LSTM}	–	–	–	–	9.60%	5.72%	***22.46%***
PV Plant #1 Target domain #4 (Partly cloudy sky)	$MSE^{\times10^{-3}}$	11.07	9.53	10.44 ± 1.40	8.69 ± 0.34	8.07 ± 0.35	8.03 ± 0.28	***6.18 ± 0.02***
	$RMSE^{\times10^{-2}}$	10.52	9.76	10.19 ± 0.70	9.32 ± 0.18	8.98 ± 0.19	8.96 ± 0.15	***7.86 ± 0.01***
	$NRMSE^{\times10^{-3}}$	2.63	2.44	2.54 ± 0.18	2.33 ± 0.04	2.24 ± 0.05	2.23 ± 0.04	***1.96 ± 0.01***
	$S_{persistence}$	–	7.24%	3.13%	11.41%	14.63%	14.85%	***25.29%***
	S_{LSTM}	–	–	–	–	3.64%	3.89%	***15.67%***
PV Plant #1 Target domain #5 (Overcast sky)	$MSE^{\times10^{-3}}$	18.97	6.72	10.13 ± 3.01	8.02 ± 0.53	6.26 ± 0.59	7.19 ± 0.93	***4.30 ± 0.08***
	$RMSE^{*\times10^{-2}}$	13.77	8.20	9.97 ± 1.47	8.95 ± 0.30	7.90 ± 0.37	8.47 ± 0.54	***6.56 ± 0.07***
	$NRMSE^{*\times10^{-3}}$	6.94	4.14	5.03 ± 0.74	4.51 ± 0.15	3.99 ± 0.19	4.27 ± 0.27	***3.31 ± 0.03***
	$S_{persistence}$	–	40.45%	27.61%	35.02%	42.61%	38.53%	***52.37%***
	S_{LSTM}	–	8.36%	–	–	11.68%	5.40%	***26.69%***
PV Plant #2 Target domain #6 (Clear sky)	$MSE^{*\times10^{-3}}$	6.77	11.82	11.75 ± 0.65	5.18 ± 0.43	5.22 ± 0.65	4.11 ± 0.52	***3.42 ± 0.07***
	$RMSE^{*\times10^{-2}}$	8.23	10.87	10.84 ± 0.30	7.19 ± 0.30	7.21 ± 0.45	6.40 ± 0.39	***5.85 ± 0.06***
	$NRMSE^{*\times10^{-3}}$	1.23	1.63	1.62 ± 0.04	1.08 ± 0.05	1.08 ± 0.07	0.96 ± 0.06	***0.87 ± 0.01***
	$S_{persistence}$	–	–	–	12.57%	12.37%	22.22%	***28.91%***
	S_{LSTM}	–	–	–	–	–	11.03%	***18.68%***
PV Plant #2 Target domain #7 (Partly cloudy sky)	$MSE^{*\times10^{-3}}$	15.26	11.63	10.22 ± 1.05	9.18 ± 0.16	9.19 ± 0.66	8.82 ± 0.36	***6.41 ± 0.06***
	$RMSE^{*\times10^{-2}}$	12.35	10.78	10.10 ± 0.53	9.58 ± 0.08	9.58 ± 0.34	9.39 ± 0.20	***8.01 ± 0.04***
	$NRMSE^{*\times10^{-3}}$	3.38	2.95	2.76 ± 0.15	2.62 ± 0.02	2.62 ± 0.09	2.57 ± 0.05	***2.19 ± 0.01***
	$S_{persistence}$	–	12.69%	18.24%	22.45%	22.44%	23.97%	***35.17%***
	S_{LSTM}	–	–	–	–	–	1.96%	***16.41%***
PV Plant #2 Target domain #8 (Overcast sky)	$MSE^{*\times10^{-3}}$	53.33	9.22	11.75 ± 1.92	9.62 ± 1.62	8.46 ± 1.10	7.84 ± 1.11	***5.43 ± 0.06***
	$RMSE^{*\times10^{-2}}$	23.09	9.60	10.81 ± 0.89	9.78 ± 0.08	9.18 ± 0.60	8.83 ± 0.64	***7.37 ± 0.04***
	$NRMSE^{*\times10^{-3}}$	13.24	5.51	6.20 ± 0.51	5.61 ± 0.48	5.26 ± 0.34	5.07 ± 0.36	***4.22 ± 0.02***
	$S_{persistence}$	–	58.43%	53.21%	57.67%	60.25%	61.75%	***68.10%***
	S_{LSTM}	–	1.80%	–	–	6.10%	9.64%	***24.65%***

thus resulting in certain difficulties in model learning; the patterns of such cases may not be learned from the adopted training dataset. Even so, the proposed C-LSTM + TL model still outperforms other forecasting models in the literature according to the obtained numerical results presented in Table 7. To improve the forecasting accuracy for such cases with high intermittency, ultra-short term (UST) PVPG forecasting is required, however, UST forecasting is not the main focus of this work.

The numerical results of PVPG forecasting for the above cases by different forecasting models are illustrated in Table 7, where the best results for each case are marked in bold. It is apparent that the proposed C-LSTM + TL model performs best in all cases with the lowest values of MSE, RMSE, and NRMSE, among all compared models, which demonstrates its great potential to provide excellent forecasting results regardless of different sky conditions. Significant improvements in forecasting accuracy (lower mean values of MSE, RMSE, and NRMSE) and stability (lower SD values of MSE, RMSE, and NRMSE) can be obtained by the C-LSTM + TL model. The forecasting skills of C-LSTM + TL can be significantly improved, up to 68.1% and 26.7%, compared with the persistence model and the standard LSTM model, respectively. The obtained results are in line with the previous discussions and confirm the effectiveness of combining TL and DL models in solving the day-ahead PVPG forecasting problem of NCPV plants. Additionally, the newly proposed C-LSTM model also outperforms the standard LSTM model with different improved rates of forecasting skill (from 2.0% to 11.0%). This proves the significance and necessity of considering the prior knowledge as physical constraints into the construction of DL models.

6. Conclusion and discussion

In this paper, combining TL and DL models, we proposed two parameter-transferring strategies and a constrained long short-term memory (C-LSTM) model, to address the hourly day-ahead PVPG forecasting problem of NCPV plants. The most appropriate TL method was determined from several experiments. The C-LSTM model was established by using the KNN algorithm to extract prior knowledge as model constraints. The obtained results verified the effectiveness of TL combined with the C-LSTM model and showed the superiority in terms of forecasting accuracy and model stability of proposed approaches over the conventional machine learning and statistical models in the literature. The forecasting skill of the combined model has improved up to 68.4% compared with the reference persistence model.

The newly proposed models in this work provide effective solutions to solve the PVPG forecasting problem of NCPV plants based on limited training data. Through C-LSTM, DL models should be not only driven by data, but also by some prior knowledge, which can assist the model to obtain better accuracy and interpretability. However, the limitations of current models are that a relatively large-size dataset used as the source domain is required to implement the proposed TL strategies. Besides, care must be taken to ensure that the input feature variables of the DL model in the source domain and target domain are the same.

In the future, the proposed TL strategies can be implemented in other DL models with more complicated structures. Some case studies in the Middle East, North China, or other regions of the world also can be carried out to verify the proposed models.

Besides, the approaches studied in this work also have the potentials to be applied to solve other limited-data-based engineering problems, such as load demand forecasting and wind power generation forecasting in future works.

CRediT authorship contribution statement

Xing Luo: Conceptualization, Investigation, Methodology, Data curation, Software, Visualization, Writing – original draft. **Dongxiao Zhang:** Conceptualization, Methodology, Supervision, Writing – review & editing, Funding acquisition. **Xu Zhu:** Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] A. Sayigh, Renewable energy - the way forward, Appl. Energy 64 (1) (1999) 15–30, https://doi.org/10.1016/S0306-2619(99)00117-8.
[2] A. Ferrara, S. Jakubek, C. Hametner, Energy management of heavy-duty fuel cell vehicles in real-world driving scenarios: robust design of strategies to maximize the hydrogen economy and system lifetime, Energy Convers. Manag. 232 (2021), 113795, https://doi.org/10.1016/j.enconman.2020.113795.
[3] M.N. Akhter, S. Mekhilef, H. Mokhlis, N. Mohamed Shah, Review on forecasting of photovoltaic power generation based on machine learning and metaheuristic techniques, IET Renew. Power Gener. 13 (7) (2019) 1009–1023, https://doi.org/10.1049/iet-rpg.2018.5649.
[4] A. Strzalka, N. Alam, E. Duminil, V. Coors, U. Eicker, Large scale integration of photovoltaics in cities, Appl. Energy 93 (2012) 413–421, https://doi.org/10.1016/j.apenergy.2011.12.033.
[5] X. Luo, D. Zhang, X. Zhu, Deep learning based forecasting of photovoltaic power generation by incorporating domain knowledge, Energy 225 (2021), 120240, https://doi.org/10.1016/j.energy.2021.120240.
[6] A. Dolara, S. Leva, G. Manzolini, Comparison of different physical models for pv power output prediction, Sol. Energy 119 (2015) 83–99, https://doi.org/10.1016/j.solener.2015.06.017.
[7] D. Koster, F. Minette, C. Braun, O. O'Nagy, Short-term and regionalized photovoltaic power forecasting enhanced by reference systems on the example of Luxembourg, Renew. Energy 132 (2019) 455–470, https://doi.org/10.1016/j.renene.2018.08.005.
[8] J. Boland, M. David, P. Lauret, Short term solar radiation forecasting: island versus continental sites, Energy 113 (2016) 186–192, https://doi.org/10.1016/j.energy.2016.06.139.
[9] S.I. Vagropoulos, G.I. Chouliaras, E.G. Kardakos, C.K. Simoglou, A.G. Bakirtzis, Comparison of sarimax, sarima, modified sarima and ann-based models for short-term pv generation forecasting, in: 2016 IEEE International Energy Conference (ENERGYCON), 2016, pp. 1–6, https://doi.org/10.1109/ENERGYCON.2016.7514029.
[10] X. Zhang, F. Fang, J. Liu, Weather-classification-mars-based photovoltaic power forecasting for energy imbalance market, IEEE Trans. Ind. Electron. 66 (11) (2019) 8692–8702, https://doi.org/10.1109/TIE.2018.2889611.
[11] H. Sheng, J. Xiao, Y. Cheng, Q. Ni, S. Wang, Short-term solar power forecasting based on weighted Gaussian process regression, IEEE Trans. Ind. Electron. 65 (1) (2018) 300–308, https://doi.org/10.1109/TIE.2017.2714127.
[12] Z. Liu, Z. Zhang, Solar forecasting by k-nearest neighbors method with weather classification and physical model, in: 2016 North American Power Symposium (NAPS), 2016, pp. 1–6, https://doi.org/10.1109/NAPS.2016.7747859.
[13] J.P. Mueller, L. Massaron, Machine Learning for Dummies, For Dummies, 2016.
[14] M. Abuella, B. Chowdhury, Solar Power Forecasting Using Artificial Neural Networks, North American Power Symposium, 2015, pp. 1–5, https://doi.org/10.1109/NAPS.2015.7335176.
[15] J. Liu, W. Fang, X. Zhang, C. Yang, An improved photovoltaic power forecasting model with the assistance of aerosol index data, IEEE Trans. Sustain. Energy 6 (2) (2015) 434–442, https://doi.org/10.1109/TSTE.2014.2381224.
[16] Y.S. Manjili, R. Vega, M.M. Jamshidi, Data-analytic-based adaptive solar energy forecasting framework, IEEE Syst. J. 12 (1) (2018) 285–296, https://doi.org/10.1109/JSYST.2017.2769483.
[17] A. Mellit, A.M. Pavan, V. Lughi, Deep learning neural networks for short-term photovoltaic power forecasting, Renew. Energy 172 (2021) 276–288, https://doi.org/10.1016/j.renene.2021.02.166.
[18] Y. Ma, Q. Lv, R. Zhang, Y. Zhang, H. Zhu, W. Yin, Short-term photovoltaic power forecasting method based on irradiance correction and error forecasting, Energy Rep. 7 (2021) 5495–5509, https://doi.org/10.1016/j.egyr.2021.08.167.
[19] G. Cervone, L.C. Harding, S. Alessandrini, L.D. Monache, Short-term photovoltaic power forecasting using artificial neural networks and an analog ensemble, Renew. Energy 108 (2017) 274–286, https://doi.org/10.1016/j.renene.2017.02.052.
[20] A.T. Eseye, J. Zhang, D. Zheng, Short-term photovoltaic solar power forecasting using a hybrid wavelet-pso-svm model based on scada and meteorological information, Renew. Energy 118 (2018) 357–367, https://doi.org/10.1016/j.renene.2017.11.011.
[21] F. Wang, Z. Zhang, C. Liu, Y. Yu, S. Pang, N. Duic, M.S. Khah, J.P. Catalao, Generative adversarial networks and convolutional neural networks based weather classification model for day ahead short-term photovoltaic power forecasting, Energy Convers. Manag. 181 (2019) 443–462, https://doi.org/10.1016/j.enconman.2018.11.074.
[22] H. Wang, Z. Lei, X. Zhang, B. Zhou, J. Peng, A review of deep learning for renewable energy forecasting, Energy Convers. Manag. 198 (2019), 111799, https://doi.org/10.1016/j.enconman.2019.111799.
[23] M. Schuster, K.K. Paliwal, Bidirectional recurrent neural networks, IEEE Trans. Signal Process. 45 (11) (1997) 2673–2681, https://doi.org/10.1109/78.650093.
[24] J. Zheng, H. Zhang, Y. Dai, B. Wang, T. Zheng, Q. Liao, Y. Liang, F. Zhang, X. Song, Time series prediction for output of multi-region solar power plants, Appl. Energy 257 (2020), 114001, https://doi.org/10.1016/j.apenergy.2019.114001.
[25] A. Gensler, J. Henze, B. Sick, N. Raabe, Deep learning for solar power forecasting - an approach using autoencoder and lstm neural networks, in: IEEE International Conference on Systems, Man, and Cybernetics, 2016, pp. 2858–2865, https://doi.org/10.1109/SMC.2016.7844673.
[26] A.N. Mohamed, M. Karar, Accurate photovoltaic power forecasting models using deep lstm-rnn, Neural Comput. Appl. (2017) 2727–2740, https://doi.org/10.1007/s00521-017-3225-z.
[27] L. Cai, J. Gu, Z. Jin, Two-layer transfer-learning-based architecture for short-term load forecasting, IEEE Trans. Ind. Inf. 16 (3) (2020) 1722–1732, https://doi.org/10.1109/TII.2019.2924326.
[28] A. van Opbroek, M.A. Ikram, M.W. Vernooij, M. de Bruijne, Transfer learning improves supervised image segmentation across imaging protocols, IEEE Trans. Med. Imag. 34 (5) (2015) 1018–1030, https://doi.org/10.1109/TMI.2014.2366792.
[29] U. Cote-Allard, C.L. Fall, A. Drouin, A. Campeau-Lecours, C. Gosselin, K. Glette, F. Laviolette, B. Gosselin, Deep learning for electromyographic hand gesture signal classification using transfer learning, IEEE Trans. Neural Syst. Rehabil. Eng. 27 (4) (2019) 760–771, https://doi.org/10.1109/TNSRE.2019.2896269.
[30] Z. Ding, M. Shao, Y. Fu, Incomplete multisource transfer learning, IEEE Transact. Neural Networks Learn. Syst. 29 (2) (2018) 310–323, https://doi.org/10.1109/TNNLS.2016.2618765.
[31] J. Asanuma, S. Doi, H. Igarashi, Transfer learning through deep learning: application to topology optimization of electric motor, IEEE Trans. Magn. 56 (3) (2020) 1–4, https://doi.org/10.1109/TMAG.2019.2956849.
[32] M. Singh, R. Tayal, M. Shrivastava, Transfer learning using bi-lstm with attention mechanism on stack exchange data, in: 2019 International Conference on Machine Learning, Big Data, Cloud and Parallel Computing (COMITCon), 2019, pp. 93–96, https://doi.org/10.1109/COMITCon.2019.8862222.
[33] J. Wang, H. Zheng, Y. Huang, X. Ding, Vehicle type recognition in surveillance images from labeled web-nature data using deep transfer learning, IEEE Trans. Intell. Transport. Syst. 19 (9) (2018) 2913–2922, https://doi.org/10.1109/TITS.2017.2765676.
[34] N. Wang, D. Zhang, H. Chang, H. Li, Deep learning of subsurface flow via theory-guided neural network, J. Hydrol. 584 (2020), 124700, https://doi.org/10.1016/j.jhydrol.2020.124700.
[35] Y. Zhang, J. Wang, K-nearest neighbors and a kernel density estimator for gefcom2014 probabilistic wind power forecasting, Int. J. Forecast. 32 (3) (2016) 1074–1080, https://doi.org/10.1016/j.ijforecast.2015.11.006.
[36] X. Luo, X. Zhu, E.G. Lim, A parametric bootstrap algorithm for cluster number determination of load pattern categorization, Energy 180 (2019) 50–60, https://doi.org/10.1016/j.energy.2019.04.089.
[37] R. Nuzzo, Scientific method: statistical errors, Nature 506 (2014) 150–152, https://doi.org/10.1038/506150a.
[38] L. von Rueden, S. Mayer, K. Beckh, B. Georgiev, S. Giesselbach, R. Heese, B. Kirsch, J. Pfrommer, A. Pick, R. Ramamurthy, M. Walczak, J. Garcke, C. Bauckhage, J. Schuecker, Informed Machine Learning – a Taxonomy and Survey of Integrating Knowledge into Learning Systems, 2019 arXiv: 1903.12394.
[39] T. Hong, P. Pinson, S. Fan, H. Zareipour, A. Troccoli, R.J. Hyndman, Probabilistic energy forecasting: global energy forecasting competition 2014 and beyond, Int. J. Forecast. 32 (3) (2016) 896–913, https://doi.org/10.1016/j.ijforecast.2016.02.001.

[40] M. Guermoui, K. Bouchouicha, N. Bailek, J.W. Boland, Forecasting intra-hour variance of photovoltaic power using a new integrated model, Energy Convers. Manag. 245 (2021), 114569, https://doi.org/10.1016/j.enconman.2021.114569.
[41] Y. Qu, J. Xu, Y. Sun, D. Liu, A temporal distributed hybrid deep learning model for day-ahead distributed pv power forecasting, Appl. Energy 304 (2021), 117704, https://doi.org/10.1016/j.apenergy.2021.117704.
[42] Y. Chu, H.T. Pedro, M. Li, C.F. Coimbra, Real-time forecasting of solar irradiance ramps with smart image processing, Sol. Energy 114 (2015) 91–104, https://doi.org/10.1016/j.solener.2015.01.024.